Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech ME V
Name of the Paper: Design of Machine Element-I
Paper Code: TME-502

Time: 3 Hour | Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

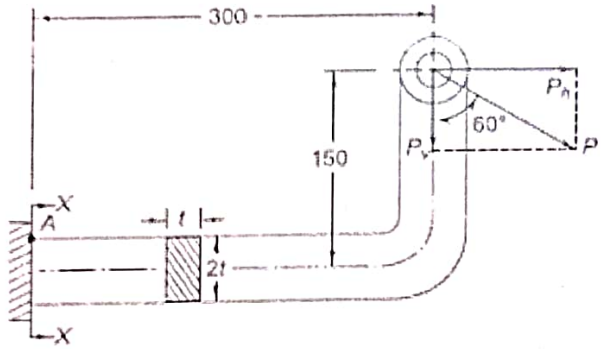
a. Explain Professional design and modern design, also write different codes(BIS) for design. CO 1

b. Write in detail about stress concentration and factor of safety. CO 1,2

c. Discuss concept of shear diagonal of distortion energy theory. CO 1, 2

Q2. (2X10=20 Marks)

a. A wall bracket (FG 200, factor of safety 3.5) with a rectangular cross-section having depth of the cross-section is twice of the width. The force 5 KN is acting on the bracket at 600 to the vertical Determine the dimensions of the cross-section of the bracket. Assume maximum normal stress theory.



CO 1,2

b. Discuss in detail maximum shear stress theory draw region of safety. CO 2

c. A machine component is made of steel having yield strength of 350MPa at a critical point of material the stress are $\sigma_{xx}= \sigma$, $\sigma_{yy}= -0.5\sigma$, $\tau_{xy}= 1.5\sigma$ determine the limiting value of σ by Rankine theory, Tresca Theory. CO 2, 3

Q3. (2X10=20 Marks)

a. Explain Cumulative fatigue damage theory. CO 2, 3

b. The work cycle of a mechanical component subjected to completely reversed bending stress consists of the following three elements ± 350 MPa for 85% of time, ±400 MPa for 12% of time, ±500 MPa for 3% of time the material of the component 50C4 (S_{ut}=660 MPa) and the corrected endurance limit of the component is 280 MPa determine the life of the component. CO 2, 3

c. A transmission shaft of cold drawn steel (S_{ut}=500 MPa, S_{yt}=300 MPa) is subjected to fluctuating torque -100 N-m to +400 N-m. the factor of safety is 2 and the expected reliability is 90% neglecting the effect of stress concentration determine the diameter of the shaft. Assume distortion energy theory. CO 2, 3

Q4. (2X10=20 Marks)

a. Determine the thickness of 120 mm wide uniform plate for safe continues operation if the plate is to be subjected to a tensile load that has a maximum value of 250 KN and minimum value of 100 KN. Endurance

limit stress 225 MPa yield point stress 280 MPa factor of safety 1.8. CO 3

b. A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque 10000 N-m the shaft is made of 45C8 S_{ut} = 700 MPa and S_{yt}=500 MPa. Take Factor of safety 6, determine the diameter of shaft. CO 3

c. Find the relation for failure due to shear stress and failure due to compressive stress in square flat key. CO 3,4

Q5. (2X10=20 Marks)

a. Explain about types of riveted joints, if two plates 16mm thick are joined by double lap joint the rivets are 25 mm in diameter find the crushing stress induced between the plates and rivet if the maximum tensile load on the joint is 48 KN. CO 3, 4, 5

b. Determine the strength of Transverse fillet weld and Parallel fillet weld. CO 4, 5

c. Explain the bolt of uniform strength in detail, an electric motor weighing of 10KN is lifted by means of an eye bolt (30C8, S_{yt} = 400 MPa). The eye bolt screwed in to the frame of motor. Factor of safety is 6. determine the size of bolt. CO 4, 5